

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Abstract

Dynamic pricing helps businesses automatically adjust product prices according to customer demand and market conditions. This project develops a Contextual Pricing Engine using the Multi-Armed Bandit (MAB) approach with Thompson Sampling. Different price levels are treated as alternative pricing strategies (arms). The system continuously learns from customer purchase behavior and updates the probability of success for each price. It balances exploration of new prices with exploitation of profitable prices while reacting to competitor price changes.

Introduction

In today's competitive online marketplace, fixed pricing often fails to maximize revenue. Customers respond differently to price changes depending on factors such as time, demand, season, and competition. This project proposes an intelligent pricing engine that continuously experiments with different prices and automatically learns the most profitable pricing strategy without requiring historical training data.

Problem Statement

Businesses often struggle to determine the optimal selling price because customer behavior changes frequently. Challenges: Maximize revenue, maintain customer conversion, respond quickly to market changes, avoid losing customers to competitors.

Objectives

- Develop a dynamic pricing simulation.
- Represent different prices as bandit arms.
- Apply Thompson Sampling.
- Update Beta distributions.
- Simulate competitor price changes.
- Compare revenue before and after dynamic pricing.

Scope

E-commerce, Online Grocery, Airline Ticket Pricing, Hotel Booking, Food Delivery, Ride Sharing.

Literature Review

Recent studies show reinforcement learning techniques are increasingly used for pricing optimization. Thompson Sampling balances exploration and exploitation while improving pricing decisions.

Dataset (Simulated)

Customer ID, Product ID, Price Offered, Competitor Price, Customer Type, Time of Day, Purchase, Revenue.

Tools

Python, Pandas, NumPy, Matplotlib, SciPy, Jupyter Notebook.

Methodology

1. Generate customer interactions.
2. Assign price arms (₹400, ₹450, ₹500, ₹550, ₹600).
3. Initialize $\text{Alpha}=1$, $\text{Beta}=1$.
4. Run Thompson Sampling.
5. Update Alpha/Beta.
6. Simulate competitor shock.
7. Repeat.

Data Preprocessing

Remove duplicates, handle missing values, encode categorical variables, normalize data.

Feature Engineering

Price Difference, Discount Percentage, Demand Level, Customer Loyalty Score, Competitor Gap, Time Category.

Algorithms Used

Multi-Armed Bandit, Thompson Sampling, Beta Distribution, Bayesian Updating.

Evaluation Metrics

Revenue, Conversion Rate, Average Selling Price, Regret, Exploration Rate, Exploitation Rate.

Expected Results

Best price identified automatically, higher revenue, improved conversion rate, adaptive pricing.

Business Insights

Returning customers accept higher prices, new customers respond to discounts, evening sales convert better.

Advantages

Automated pricing, continuous learning, competitor response, improved revenue.

Limitations

Needs continuous interaction data and realistic simulations.

Conclusion

Thompson Sampling enables intelligent dynamic pricing by continuously learning customer responses and adapting prices.